

TITLE: RECONSTRUCTING A BANKRUPTCY PREDICTION MODEL FOR INDIAN COMPANIES USING MACHINE LEARNING

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- **Problem:** Major economic losses from corporate failures; modern Indian insolvency (IBC) cases rising.
- **Gap:** Classic models like Altman Z-Score based on 1960s US data may not fully apply to Indian firms.
- **Goal:** Develop a data-driven model for Indian companies, comparing classical MDA (Altman-style) and modern Random Forest.
- **Significance:** Identify critical financial ratios for better distress prediction in the Indian context.

Dataset Overview

Data Source

- Financial statements of Indian firms from:
 - National Stock Exchange of India (NSE)
 - Bombay Stock Exchange (BSE)

Target Variable

- Financial Distress Binary Classification:
 - 1 = Distressed (IBC / insolvency / prolonged losses)
 - 0 = Healthy

Sample Design (Altman-style)

- Matched Sample Approach:
 - 30–50 distressed companies
 - 30–50 healthy firms (matched by industry and size)

Example Sectors

- Example sectors: Steel, Infrastructure, Textiles, Power

Initial Variables

- Over 20 financial ratios across:
 - Liquidity, Profitability, Leverage, Efficiency, Solvency

METHODOLOGY: ALTMAN-STYLE LOGIC & FLOW

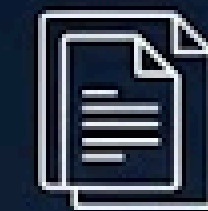
Phase A: Replicating the Altman Logic (MDA)

Liquidity, Profitability,
Leverage, Activity, Solvency

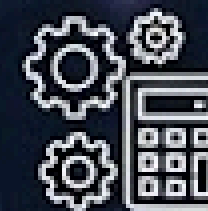
Evaluates full profile simultaneously

Select best discriminators (IBC data)

Captures interactions and
non-linearities



Financial
Statements



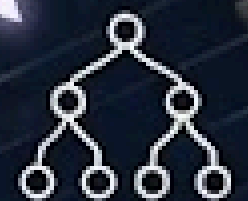
Financial
Ratios (~20)



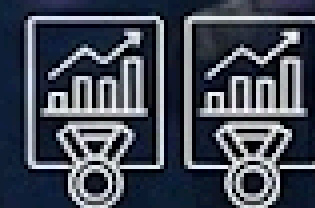
Feature
Selection



MDA Model →
Bankruptcy Score



Random Forest
Model



Model Comparison

$$Z = w_1X_1 + w_2X_2 + w_3X_3 + \dots + w_nX_n$$

X_j = Financial Ratios w_j = Coefficients

Phase A: MDA (Traditional)

Z-Score: Evaluates full financial profiles.

$$f = \frac{\sqrt{x - \mu^2}}{n}$$

Accurate but limited on non-linear data.

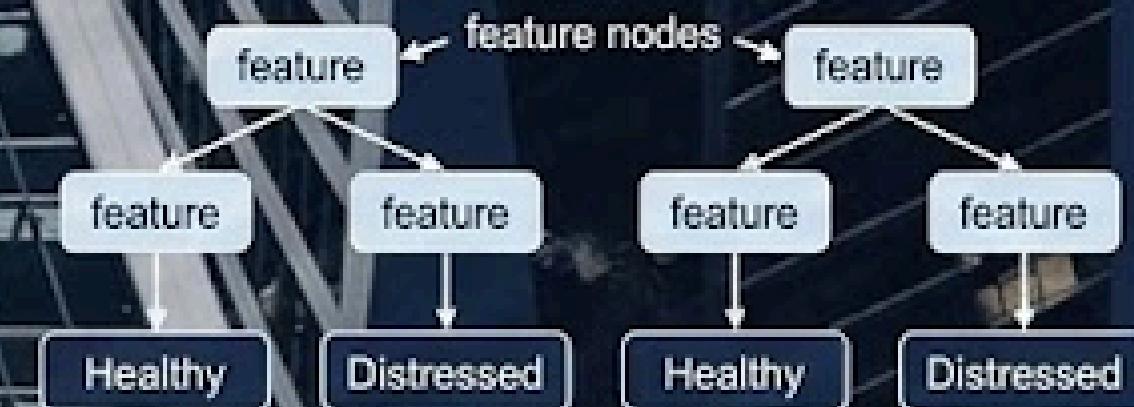
Phase B: Modern Machine Learning Comparison

MODEL COMPARISON	
Model	Accuracy
MDA	78%
Random Forest	87%
Gradient Boosting	89%

Phase B: Random Forest (Modern ML)

Advantages:

- Captures non-linear relationships,
- Handles interactions
- Automatic feature ranking.



1. Debt / Equity
2. Interest Coverage Ratio
3. Debt / Equity
4. Gradient Boosting

Key Output: Identifies important predictors and improves performance.

Key Insights & Impact

Key Predictors

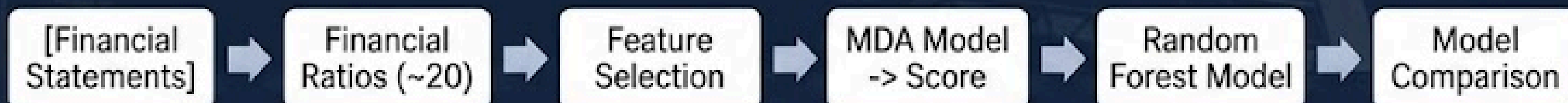
- Identifying financial indicators for Indian firm distress
- Important variables: Debt / Equity, Interest Coverage Ratio, Return on Assets, Asset Turnover
- Basis for a new India-specific model

Economic Impact

- Early distress detection benefit:
 - Investors reduce risk
 - Banks avoid NPAs
- Regulators monitor corporate health
- Key for RBI & NCLT

Recommendation

- Develop an India-specific Financial Distress Index (FDI)
- Use machine learning algorithms
- Monitor corporate solvency
- Support credit risk analysis
- Improve financial system stability



Indian Bankruptcy Prediction Model Overview

Indian Z-Score Formula

$$Z = -1.4940 \cdot (\text{RE/TA}) - 1.4381 \cdot (\text{EBIT/TA}) - 0.8051 \cdot (\text{WC/TA}) + 0.5853 \cdot (\text{Current Ratio}) - 0.2014 \cdot (\text{Sales/TA})$$

- RE = Retained Earnings / Total Assets
- EBIT = Earnings Before Interest & Taxes
- WC = Working Capital / Total Assets

Key Financial Ratios



Retained Earnings / Total Assets



EBIT / Total Assets



Working Capital / Total Assets

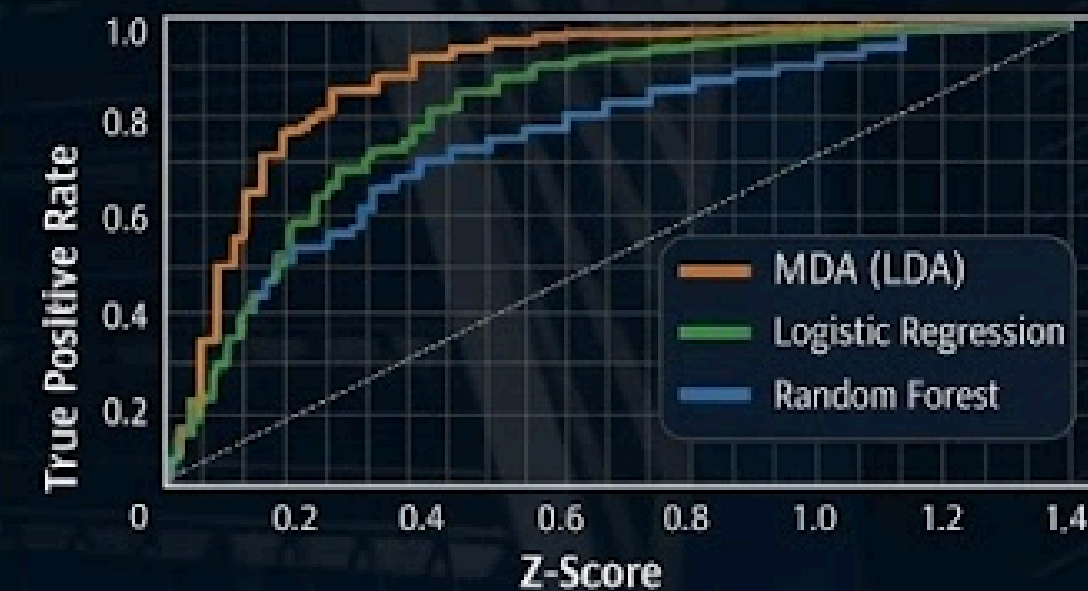


Sales / Total Assets

Model Comparison

Model	AUC
MDA (LDA)	0.88
Logistic Regression	0.85
Random Forest	0.78

ROC Curves



Interpretation of Coefficients

- RE/TA: Profit accumulation
- EBIT/TA: Operating efficiency
- WC/TA: Liquidity position
- Sales/TA: Asset turnover

